Evolving a Printable Titanium Karambit under Ergonomic, Edge and Serration Constraints

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***Abstract*—We report the best individual of the G3 v3.0.0 cohort of record, a one-piece Ti-6Al-4V karambit for laser powder-bed fusion with a hollow, lattice-filled grip and blade core. No person designed it: 24 independent CVT-MAP-Elites islands ran 2,500 evaluations each from scrambled Sobol samples, and the design space itself carries the rules a good edge must follow. Both cutting edges have one primary grind of 20.5–20.8° and 20.6–21.6° per side, measured normal to the edge and carried to within 3.0 and 3.1 mm of the point, and both edges are serrated (3 and 5 teeth; the weakest tooth has a safety factor of 2.83 under a 180 N snag). The design scores 80.55 under W_ERGOLIGHT_v1 draft d with an ergonomic multiplier of 0.988 and weighs 59.3 g as printed, which is above the 50 g goal. The 0.4 mm search grid undercounted the mass of the search's own leader (51.3 g against 75.2 g at 0.2 mm and 74.6 g at 0.15 mm), so the ranking of record is measured on the 0.2 mm grid of the print file. Against a safety-factor target of 1, its solid tip (0.64) is below the target and its grip under the spine load (0.74 by voxel FE) is below the target. We describe the representation, objective, search and measurement chain, the evolution that produced the design, and every limit of the result.**

***Index Terms*—Evolutionary design, quality diversity, MAP-Elites, additive manufacturing, Ti-6Al-4V, finite elements, knife design.**

1 Introduction

A karambit is a curved, ring-retained utility and defensive knife gripped in a closed fist with the index or little finger seated through a rear retention ring. Fabricating a one-piece karambit in Ti-6Al-4V by laser powder-bed fusion (L-PBF) opens an expressive three-dimensional design space: the grip and blade core can be hollowed into a skin over a triply periodic minimal-surface (TPMS) lattice with integral powder-evacuation ports, while variable-thickness ribs, finger grooves and dual-edge bevels are built directly into the single monolithic part. Yet unrestricted evolutionary search in three dimensions also creates severe specification traps. A coarse voxel grid can hide mass or bridge thin gaps; a beam or plate surrogate can credit interior material that is absent from the printed build; a blade profile can leave an unground shoulder where a sparse station check does not look; and deep, narrow serration gullets can inflate apparent edge length while leaving brittle needle teeth that snap under an oblique snag.

Generation 3 (G3) of the Jarambit project addresses these failure modes by formulating the whole knife—retention ring, ergonomic grip, guard, hollow lattice interior, and a double-edged, double-serrated claw blade—inside a unified constructive decoder and multi-fidelity finite-element evaluation pipeline. This paper reports the v3.0.0 release and the best individual of its cohort of record, `hz4i013_g00995_d206e3c9` (commit `80e669e5b558`, decoder hash `a203278d80d2`).

Three architectural principles govern the v3.0.0 release:

1) *Evolution designs the knife from scratch.* Every island in the cohort is initialised exclusively from scrambled low-discrepancy Sobol sequences (g3_genome.rs). No hand-tuned seed, template CAD model or post-hoc manual edit enters the population. Engineering laws are encoded either as invariant geometric mappings inside the decoder or as measured physical responses in the objective.
2) *Edge and serration mechanics are intrinsic to the decoder.* Rather than treating sharpening and serrations as a post-process applied after structural optimisation, decoder revision r2 enforces five primary-grind invariants (EDGE-1..5) and four serration-integrity laws (SERR-1..4) directly on the signed distance field (g3_sdf.rs; Section 2). Both cutting edges carry a single primary bevel at a titanium-appropriate angle (17.5–22.5° per side) measured normal to the true serrated edge curve, run unbroken to the piercing point, and carry scalloped serrations whose individual tooth roots are checked against combined in-plane snag (180 N) and cross-blade pry (60 N) loads (serration_tooth_strength).
3) *Every reported quantity traces to a single committed measurement receipt.* All scores, dimensions, safety factors, convergence checks and rankings in this paper and in the repository README.md are generated automatically from evidence/g3/v3.0.0/release.json via g3_release_tex.rs and gen_readme.py. Analytic Tier-0 screening estimates never appear in reported results; every quoted value comes from voxel-SDF geometry, plate and solid finite elements, and an explicit hex8 voxel finite-element certification of the printed part.

The remainder of this paper details the 179-gene representation and edge/serration decoder laws (Section 2), the veto-free multiplicative-ergonomics objective W_ERGOLIGHT_v1 draft d (Section 3), the 24-island CVT-MAP-Elites search and three-tier evaluation cascade (Section 4), the multi-grid and voxel finite-element verification chain (Section 5), the evolutionary trajectory and best individual `hz4i013_g00995_d206e3c9` (Section 6), the explicit physical and modelling limits of the result (Section 7), and the complete reproduction protocol (Section 8).

Release v3.0.0, decoder commit `80e669e5b558` (decoder hash `a203278d80d2`). All figures and numbers in this paper are generated from the release receipt evidence/g3/v3.0.0/release.json. Status: computational design; not built, printed or physically tested.

2 Design space

2.1 Genome, constructive SDF and hollow lattice AM

A G3.0 genome is a vector of 179 real genes $\mathbf{u} \in [0,1]^{179}$ under schema G3.0 (g3_genome.rs). The deterministic mapping G3Genome::decode transforms $\mathbf{u}$ into physical parameters (G3 Params) that drive a constructive signed distance field (SDF) of the complete knife (g3_sdf.rs):

- **Retention ring and hand-zone grip:** a finger ring with a fixed 24.9 mm internal bore diameter and an 8.0 mm nominal radial lip (lip_w $\in [6.5, 10.0]$ mm), coupled through a filleted junction to an ergonomic palm swell with four anatomical finger saddles, variable section aspect ratio, and a blended procedural surface texture (knurl, Turing reaction-diffusion, Voronoi, longitudinal ridge and CPPN bases).
- **Guard and curved claw blade:** an integral finger guard and a curved karambit blade whose straight-line chordal reach from the guard is hard-capped at 85.725 mm (3.375 in, BLADE_REACH_CAP_MM) with a spine-to-tip hook drop and a five-station thickness spine tapering smoothly toward the point.
- **Hollow TPMS interior and powder drainage:** inside the exterior shell wall (t_wall_grip, t_skin_blade), the grip, guard and blade core are hollowed and infilled with a continuous blend of Gyroid and Schwarz-D triply periodic minimal-surface (TPMS) lattices (tpms_val, tpms_level_for_rho) reinforced by internal stiffening ribs and self-supporting longitudinal corbels. To guarantee manufacturability in L-PBF Ti-6Al-4V, the voxelised build undergoes variable-radius 2D-z morphological closing and opening (morph_close_variable_2d_z, morph_open_variable_2d_z), a 26-connectivity solid flood fill (keep_connected_solid_26) that deletes any unanchored island, and a 6-connectivity void flood fill from the powder-evacuation ports so that trapped internal cavities are penalized.

Decoder revision r2 is locked by the SHA-256 digest `a203278d80d2` computed over the canonical decoder source files (g3_decoder_sha256 in g3_genome.rs); any archive or receipt whose recorded decoder hash differs fails admission.

2.2 Primary-grind edge laws (EDGE-1..5)

To ensure that every decoded candidate is a true two-edged cutting tool rather than a blunt structural strut with a cosmetic chamfer, decoder revision r2 enforces five geometric and measurement rules across both the concave inner edge (Edge A) and the convex spine edge (Edge B):

1) **EDGE-1 (Titanium-appropriate primary bevel angle):** the per-side bevel angle genes angle_in and angle_sp are bounded to $[17.5°, 22.5°]$ (35°–45° inclusive apex angle, centred on 20° per side for hard-use Ti-6Al-4V).
2) **EDGE-2 (True distance to the serrated edge curve):** at every planform point (x, y) on the blade, the local bevel depth is evaluated from the exact Euclidean distance $d_{\text{edge}}(x, y)$ to the serrated edge polyline (BladeProfile2D in g3_sdf.rs), rather than from a longitudinal chord projection. Consequently, the per-side grind angle measured along the true edge normal remains constant around the curved claw and around every serration scallop.
3) **EDGE-3 (Full-height primary grind without micro-bevel or shoulder):** the V-grind thickness ceiling $t_{\text{grind}}(d) = t_{\text{apex}} + 2\,d\tan\theta_{\text{edge}}$ is carried outward from the apex until it meets the structural core thickness of the blade, with no secondary micro-bevel clamp and no step shoulder behind the edge.
4) **EDGE-4 (Full-length sharpened reach with bounded guard plunge):** both edges run unbroken all the way to the piercing point ($s_1 = 1.0$). Only a short plunge transition next to the guard (s0_in, s0_sp $\in [0.05, 0.12]$ of the edge arc) remains unsharpened, subject to a 101.6 mm (4.0 in) maximum sharpened arc length per edge.
5) **EDGE-5 (True edge-normal section audit):** the sharpness evaluator (evaluate_gate_s_sharpness in g3_qd.rs) samples the 3D blade thickness along the true outward normal $\mathbf{n}_{\text{edge}}$ over the first 2.0 mm behind the apex across both edges, so any blunt station or recessed shoulder directly degrades the sharpness score.

2.3 Double-serrated edge and tooth-strength laws (SERR-1..4)

Both cutting edges are required to carry functional, break-resistant serrations governed by four rules:

1) **SERR-1 (Mandatory dual-edge serration depth):** the serration depth genes are bounded to $d_{\text{in}} \in [0.8, 2.2]$ mm on the inner edge and $d_{\text{sp}} \in [0.8, 2.0]$ mm on the spine edge, eliminating smooth-edge degenerate solutions.
2) **SERR-2 (Minimum tooth census):** each serrated span is required to accommodate at least three full-depth teeth (serration_census in g3_genome.rs).
3) **SERR-3 (Plunge clearance and plain piercing point):** on each edge, the serrated span starts clear of the guard plunge ($s_{0,\text{ser}} \geq s_{\text{plunge}} + 0.06$ on Edge A; +0.08 on Edge B) and ends at least 18.0 mm of arc length before the tip, leaving a smooth, continuous plain-edge kissaki point for piercing and tip prying.
4) **SERR-4 (Bounded aspect ratio and as-built tooth-root stress verification):** to prevent narrow, fragile teeth, the tooth pitch is bounded to $p_s \in [5.0, 9.0]$ mm and the gullet depth is clamped to at most one quarter of the pitch ($d_s \leq 0.25\,p_s$, bounding the mean flank angle to at most 26.6°). Every individual tooth on both edges is then verified structurally from the decoded 2D/3D blade geometry (serration_gullets in g3_sdf.rs and serration_tooth_strength in g3_qd.rs). For each tooth bounded by adjacent gullet bottoms $\mathbf{g}_0, \mathbf{g}_1$ with root chord $c = \|\mathbf{g}_1 - \mathbf{g}_0\|$, tooth height h_t, gullet curvature radius ρ, and five-point Simpson-integrated root thickness $\bar{t}$ and out-of-plane section modulus Z_{lat}, two load cases are evaluated against $\sigma_{\text{allow}} = 795$ MPa:

$$\sigma_{\text{snag}} = K_t \frac{F_{\text{snag}}}{\bar{t}\,c} + K_t \frac{6\,F_{\text{snag}}\,h_t}{\bar{t}\,c^2}, \qquad \sigma_{\text{pry}} = \frac{F_{\text{pry}}\,h_t}{Z_{\text{lat}}}, \quad (1)$$

where $F_{\text{snag}} = 180$ N acts along the edge at the tooth tip with Inglis notch factor $K_t = 1 + 2\sqrt{d_s/\rho}$ [6], and $F_{\text{pry}} = 60$ N acts perpendicular to the blade plane at the tooth tip. The decoder regression suite verifies that

Table 1 | Weights w_i and ergonomic exponents β_k of W_ERGOLIGHT_v1 (draft d)

Additive term s_i	w_i	Ergonomic factor e_k	β_k
Mass (linear to 25 g)	0.3700	Ring lip (lip_w, bore)	0.18
Solid tip 3D (TET10)	0.0700	Grip section (perim., H/T)	0.18
Gate-S edge sharpness	0.0500	Hand-contact edge radii	0.18
Gate-S point profile	0.0500	Finger groove indexing	0.15
AM build & drainage	0.0500	CoM & spin inertia	0.10
CAD watertight census	0.0400	Grip chord length L_g	0.08
Karambit ring & claw	0.0400	Guard height h_{guard}	0.08
Form fairness	0.0400	Grip texture amp./crest	0.05
Hiker lateral (180 N)	0.0305		
Hunter lateral (260 N)	0.0305		
Yield sequence (≥ 1.2)	0.0305		
Ring back (30 N, SF 2)	0.0261		
Ring bearing (300 N, SF 2)	0.0218		
Blade body twist	0.0218		
Blade tip twist	0.0218		
3-level plate GCI	0.0200		
Strict lateral (260 N)	0.0174		
Blade buckling λ	0.0174		
Human-max twist/spine	0.0174		
Goodman L7 fatigue	0.0174		
Hollow grip SF_{grip}	0.0174		

every tooth across 60 Sobol-sampled genomes achieves $\text{SF}_{\text{tooth}} = \sigma_{\text{allow}} / \max(\sigma_{\text{snag}}, \sigma_{\text{pry}}) \geq 1.0$.

3 Objective

Candidates are ranked by the veto-free continuous objective W_ERGOLIGHT_v1 (draft d, requirement set RS-karambit-v3; ergolight.rs):

$$\text{score}_{100} = 100\, E \sum_{i=1}^{21} w_i\, s_i, \qquad E = \prod_{k=1}^{8} e_k^{\beta_k}, \tag{2}$$

where $s_i \in [0,1]$ are normalized physical term scores with weights $\sum_{i=1}^{21} w_i = 1$, and $E \in [0.3, 1.0]$ is a geometric-mean ergonomic multiplier over eight hand-zone compatibility factors $e_k \in [0.3, 1.0]$ with exponents $\sum_{k=1}^{8} \beta_k = 1$. Because E multiplies the entire structural and mass sum rather than contributing an additive bonus, a candidate cannot trade hand comfort for grams: any violation of the anatomical target bands scales down every point earned elsewhere. Conversely, once all eight hand-zone measurements lie within their target plateau bands, $e_k = 1$ and $E = 1$, so ergonomics never distorts mass or strength trade-offs among already comfortable knives.

Table 1 lists the 21 additive weights w_i and 8 ergonomic exponents β_k. Mass carries the single largest weight ($w_{\text{mass}} = 0.370$), mapped linearly from 0 at 160 g to 1 at 25 g so lighter designs are always rewarded. In draft d, the 3D solid-tip pry term carries $w_{\text{tip}} = 0.070$ to drive tip-root reinforcement under the 1.82 N·m tip pry, while Gate-S edge sharpness, Gate-S point profile, and AM buildability carry 0.050 each. Every structural safety-factor ratio $v = \text{SF}/\text{SF}_{\text{req}}$ is mapped through a smooth knee function (sfk in ergolight.rs),

$$\text{sfk}(v) = 0.94\, \min(\max(v, 0), 1) + 0.06\, (1 - e^{-\max(0, v-1)/0.15}), \tag{3}$$

which awards 94 % of the term credit continuously up to first-yield adequacy ($v = 1$) and saturates the remaining 6 % exponentially above $v = 1$ to reward margin against mesh and manufacturing variation without encouraging over-built mass. No term acts as a hard reject gate during evolution, preserving smooth fitness gradients across the archive.

4 Search

4.1 CVT-MAP-Elites archive and variation operators

Each island evolves a centroidal Voronoi tessellation MAP-Elites (CVT-MAP-Elites) archive [1], [2] of $K = 1{,}024$ niches over an eight-dimensional measured behaviour descriptor space $\mathbf{b} \in [0,1]^8$ (G3Descriptors in g3_qd.rs): (1) normalized hook drop, (2) inner edge turn angle, (3) silhouette perimeter-to-area complexity, (4) grip section roundness T/H, (5) finger groove articulation depth, (6) internal TPMS porosity fraction, (7) procedural surface texture energy, and (8) normalized mass. Niche centroids are generated deterministically by Lloyd's relaxation over low-discrepancy samples so every island shares an identical 1,024-cell behavioural partition.

Every island is initialised from 64 Owen-scrambled Joe–Kuo Sobol points [3], [4] in $[0,1]^{179}$ (sobol_owen_point_179d in g3_genome.rs), with no hand-crafted or previously evolved seeds. Subsequent evaluations breed one child per step from two archived parents via directional iso-line variation (iso_dd _offspring, $\sigma_{\text{iso}} = 0.018$, $\sigma_{\text{dir}} = 0.085$) combined with sparse single-gene jump perturbations. Parent selection draws 40 % of pairs from the top-scoring elites in the archive and 60 % from a curiosity-weighted three-way tournament that up-weights niches whose offspring recently produced archive admissions. Under the tier-aware admission rule (G3CvtArchive::try_insert), a candidate claims its nearest Voronoi cell if the cell is empty, if the candidate was evaluated at a strictly higher fidelity tier than the incumbent, or if it achieves a higher score at the same tier.

4.2 Three-tier multi-fidelity evaluation cascade

To focus finite-element compute on promising morphologies, every candidate passes through a three-tier cascade (g3_qd.rs):

1) **Tier 0 (Analytic geometry & beam screen):** decodes the 2D blade and grip profiles, verifies geometric admissibility, and evaluates closed-form beam, lateral-torsional buckling (ltb_sum) and twist surrogates in a few milliseconds. Only candidates scoring at least 62 advance to Tier 1.
2) **Tier 1 (3D voxel SDF & Reissner–Mindlin plate FE):** voxelises the full 3D knife and internal TPMS lattice at pitch $h = 0.4$ mm (build_g3_grids), measures exact voxel mass, centre of mass, spin inertia and 6-connected powder drainage, and solves 3-level Reissner–Mindlin MITC3 triangular shell/plate finite-element models of the curved blade and retention ring under the lateral (180 N, 260 N), ring bearing/back (300 N, 30 N), human-maximum twist (16.2 N·m) and spine (3,427 N) load cases (evaluate_tier1_sdf_and_plate). Candidates scoring at least 71 at Tier 1 advance to Tier 2.
3) **Tier 2 (Watertight mesh census & 3D TET10 solid-tip FE):** extracts the marching-cubes surface mesh [5] with fail-closed grid-shift recovery (extract_watertight_mes h_fail_closed: zero boundary edges, zero non-manifold

edges) and solves a 3D quadratic tetrahedral (TET10) solid finite-element model of the distal 30 mm of the blade under a 1.82 N·m tip pry at a converged planform triangle area limit of TIP_FE_MAX_AREA_MM2 = 0.15 mm^2 (evaluate_tier2_elites).

Only Tier-1 and Tier-2 evaluations constitute physical measurements; Tier-0 surrogates are used solely to filter poor offspring and are never reported as results.

4.3 Independent islands and RNG-isolated trajectory logging

The cohort runs 24 independent islands in parallel, each pinned to a single worker thread with a distinct deterministic seed derived from host and island indices (g3_evolve.rs). Every 50 evaluations (--log-every 50), run_g3_cvt_map_elites_logged appends a G3HistoryPoint record to history.jsonl capturing the evaluation count, filled niches, Tier-1 and Tier-2 elite counts, and the highest measured (Tier-1/Tier-2) score and mass, together with the full 179-gene CSV row at power-of-two checkpoints and whenever a new measured best is discovered. Because trajectory logging reads archive state without drawing from the pseudo-random number generator, a logged evolution run produces a bit-for-bit identical archive to an unlogged run.

5 Measurement and verification

5.1 Platform and grid

The cohort ran on xbabe4 (arm64 macOS). Floating-point libraries differ between platforms, so the final ranking was repeated on Linux x86_64 (xbabe2), Tier 2 at h 0.2 mm, which is the platform of record for every number below; the winner scored 80.55 on the search host and 80.55 on the platform of record. The search grid is h = 0.4 mm and the ranking grid h = 0.2 mm (Section 6). Mass is reported from the watertight 0.2 mm mesh of the printed part (13,385 mm^3 at 0.00443 g/mm^3, 59.3 g); the ranking counted 57.0 g on its voxel grid and the search 56.0 g.

5.2 Solid tip

The point is checked by a solid TET10 model of the last 30 mm of the blade under a 1.82 N·m pry, reading the peak nodal von Mises stress 4–15 mm from the point. Its planform mesh was refined until it converged on sharp points: for this design the safety factor is 0.585, 0.638 and 0.611 at maximum triangle areas of 0.4, 0.15 (production) and 0.08 mm^2.

5.3 Voxel finite-element certification

The plate and beam models of the search are fast surrogates. For the winner we also solved the printed geometry itself: a hex8 linear-elastic model on its AM-build voxels at h = 0.4 mm (skin, lattice, ribs and ports as built), in two zones, the grip and the blade root. Each zone is clamped at one end and loaded at the other with the wrench of three cases: 260 N lateral at the tip, the 16.2 N·m human-maximum twist, and 3427 N on the spine at the blade centroid. We report the 99.9th-percentile von Mises stress of each zone against an allowable of 674 MPa, not a single-element peak, because the voxel staircase concentrates stress artificially at re-entrant corners. Table 2 compares the two.

Table 2 | Safety factors of the winner: search surrogates and voxel finite elements of the printed geometry

Load case	Grip formula	Grip, voxel FE	Root, voxel FE
Lateral 260 N	5.07	3.55	2.08
Twist 16.2 N·m	12.69	5.50	1.69
Spine 3427 N	0.90	0.74	1.29

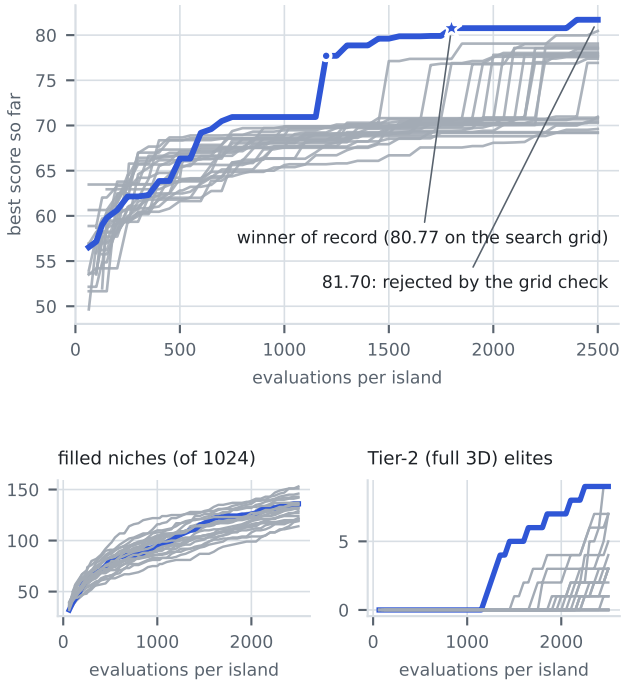


Figure 1 | Top: best measured score so far of each island (the winning island in colour; the dot marks its first full-3D best). Bottom: filled niches and Tier-2 elites per island.

6 Results

6.1 The evolution

The cohort ran 24 islands of 2,500 evaluations each, 60,000 in all, from 2026-10-26T11:03:49Z to 2026-10-26T12:32:55Z UTC, and kept 3,177 elites, 68 of them at Tier 2. Fig. 1 shows the best measured score of every island against evaluations, and how the archives filled. The winner came from island island13, which started from a best measured score of 56.54 after its Sobol initialisation. Fig. 2 shows that island's best design at each logged checkpoint, from the same camera.

6.2 Choosing the winner on the print grid

The search measures every design on an h = 0.4 mm grid, and that grid can undercount mass. The search's own leader, `hz4i013_g00840_2a44149e` (Tier-2 score 81.53 at h = 0.4 mm), weighs 51.3 g at h = 0.4 mm but 75.2 g at h = 0.2 mm and 74.6 g at h = 0.15 mm, and its Tier-1 score falls from 75.06 to 66.65 (Table 3). The two finest grids agree within 3.3 % for every design, so every number of record is measured at h = 0.2 mm, the grid of the print file. The final ranking ordered every elite by its record from the run, re-evaluated the top 32 at Tier



eval 64 56.54, 72.1 g | eval 128 59.01, 65.3 g | eval 256 62.15, 68.0 g | eval 512 66.34, 59.9 g | eval 1,024 70.94, 62.4 g | eval 2,048 80.77, 56.0 g **winner of record** | eval 2,500 81.70, 51.3 g **rejected by the grid check**

Figure 2 | Best design of island island13 at each logged checkpoint: evaluation, best measured score and mass on the search grid. Same camera, light and material for every frame; geometry is the exported mesh.

Table 3 | Grid check: mass (g) of the top 8 of the search-grid ranking at five voxel pitches h (mm), Tier 1

Design	0.4	0.3	0.25	0.2	0.15
`hz4i013_g00840_2a44149e`	51.3	55.6	54.9	75.2	74.6
`hz4i013_g00995_d206e3c9`	56.0	60.0	59.0	57.0	57.5
`hz4i002_g00995_6f4643f1`	58.8	59.6	58.9	58.7	58.5
`hz4i013_g00345_be45c4bc`	55.3	59.7	56.4	56.7	57.1
`hz4i002_g00479_76f5e0ac`	58.0	61.7	58.2	59.5	58.9
`hz4i013_g00555_5cd1f106`	50.6	54.1	54.4	52.1	53.1
`hz4i005_g00345_c7922cf1`	56.8	61.0	57.6	58.5	59.7
`hz4i002_g00785_6b2ff4aa`	59.1	62.0	58.8	61.8	59.8

Table 4 | Final ranking: Tier 2 at $h = 0.2$ mm on the platform of record

Rank	Design	Score	Mass (g)
1	`hz4i013_g00995_d206e3c9`	80.55	57.0
2	`hz4i002_g00995_6f4643f1`	80.44	58.7
3	`hz4i013_g00345_be45c4bc`	79.51	56.7
4	`hz4i002_g00479_76f5e0ac`	79.39	59.5
5	`hz4i013_g00555_5cd1f106`	79.27	52.1
6	`hz4i004_g00501_cb3c9e4e`	79.24	57.3
7	`hz4i002_g00905_b274d4d9`	78.94	59.6
8	`hz4i007_g00005_98a9344b`	78.77	53.7

2 at $h = 0.2$ mm, and the top 8 again on the platform of record (Table 4).

6.3 The best individual

The best individual, `hz4i013_g00995_d206e3c9` (genome hash `d206e3c9f4c3`), scores 80.55 with a weighted base of 0.815 and an ergonomic multiplier of 0.988. Its grip is 64.9 mm long with a swell perimeter of 64.4 mm and a height-to-width ratio of 1.59; the guard is 11.4 mm high and the reach of the blade from the guard is 69.3 mm. As printed it weighs 59.3 g. Fig. 3 shows it; Table 5 lists its measurements.

6.4 Edges and serrations

Fig. 4 plots the blade section at four stations of each edge, measured along the true edge normal with the exact decoder law. Every station of an edge lies on one line: edge A at 20.5–20.8° per side and edge B at 20.6–21.6°, with no slope steeper than 20.9° and 21.9° within 2 mm of the edge. Both edges are serrated over the rear part of the blade, and the grinds run on through the plain point region to 3.0 and 3.1 mm from the point. Under the 180 N snag and 60 N pry the weakest tooth has a safety factor of 2.83, which meets the target of 1.

Table 5 | Measurements of the best individual (platform of record)

Measure	Value
Mass, printed (watertight mesh)	59.3 g
Edge A per-side angle, apex / bevel	20.5–20.8/ 19.5–20.8°
Edge B per-side angle, apex / bevel	20.6–21.6/ 18.7–21.7°
Grind ends from the point, A / B	3.0 / 3.1 mm
Unsharpened plunge, A / B	9.7 / 11.9 mm
Serration depth, A / B	1.63 / 1.41 mm
Teeth, A / B	3 / 5
Weakest tooth SF, A / B	3.47 / 2.83
Solid tip SF (1.82 N·m)	0.64
Lateral capacity (plate FE)	337 N
Grip SF (formula)	0.73
Human-max twist SF (plate)	1.09
Spine 3427 N SF (plate)	1.44
Ring bearing / back capacity	8,677 / 1,686 N
Fatigue SF / yield-sequence ratio	0.94 / 0.56
Buckling factor λ	11.2

7 Limits of the result

What this result is not

It is a computational design. Nothing has been printed, heat-treated, sharpened or tested, and no safety factor here is a qualification.

- **Strength against the targets.** Against a target of 1, the solid tip (0.64) is below the target; the grip under the spine load (formula 0.73, voxel FE 0.74) is below the target; the human-maximum twist (plate 1.09, voxel FE 1.69) meets the target; fatigue (0.94) is below the target; and the yield-sequence ratio (0.56 against 1.2) is below the target. Lateral capacity meets the target and the ring meets the target.
- **Mass.** The design weighs 59.3 g, which is above the 50 g goal.
- **Assumed loads.** The tooth snag and pry loads are conservative research assumptions, not measured use loads. The buckling factor is an analytic lateral-torsional estimate [7], not an eigenvalue solve.
- **Models.** Voxel finite elements resolve the printed geometry only to $h = 0.4$ mm and report percentiles, not peaks; the plate and beam surrogates of the search differ from them (Table 2). Surfaces come from marching cubes on the voxel SDF [5], except the blade, which is re-meshed at 0.05 mm from the exact decoder law.
- **Coarse search grid.** The archive was selected at $h = 0.4$ mm, which undercounts the mass of some designs (Table 3); only the final ranking is measured at $h = 0.2$



Figure 3 | `hz4i013_g00995_d206e3c9`. Top: the knife, and an x-ray with every internal cavity in amber. Bottom: a section across the blade 30 mm from the point, the point, and the spine edge with its serrations. Cycles renders of the exported meshes; the blade is meshed at 0.05 mm from the exact decoder law, the rest at 0.2 mm voxels.

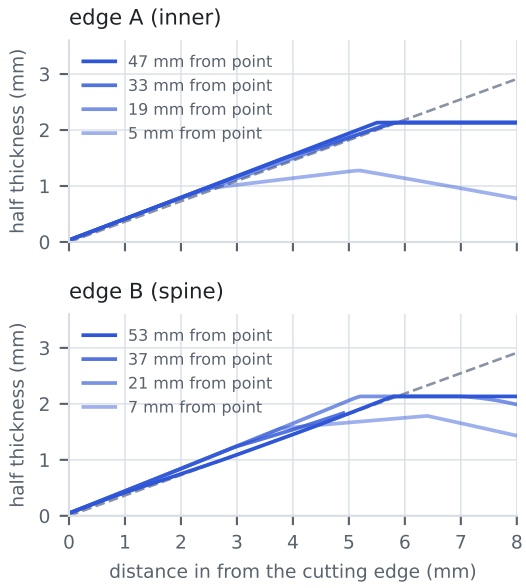


Figure 4 | Upper half of the blade section at four stations of each edge, true scale, against a 20° per-side grind (dashed). Profiles stop where a lightening slot begins.

mm, so designs that the coarse grid scored low were never re-checked.

- **One shortened cohort.** The release cohort is 2,500 evaluations per island on one host; another seed or a longer run may find a better design.

Table 6 | CAD files of the best individual (binary STL, millimetres; each file name starts with the design identifier)

File	Triangles	MiB	Where
`AMbuild_mm.stl`	1,001,372	48	git and release
`envelope_mm.stl`	911,132	43	git and release
`AMbuild_h0.2_mm.stl`	5,373,548	256	release
`envelope_h0.2_mm.stl`	2,063,960	98	release
`envelope_fineblade_mm.stl`	3,843,106	183	release

8 Reproducibility

The release is commit `80e669e5b558` with decoder hash `a203278d80d2`. The winner's 179-gene row is evidence/g3/v3.0.0/winner_genome.csv; every island's logged history, the ranking records, the edge profile, the blade facts and the finite-element results sit next to it under evidence/g3/v3.0.0/, listed with their SHA-256 in SHA256SUMS. The receipt release.json is the single source of this paper and of the repository README: g3-release-tex turns it into the macros used here, and ops/ci/gen_readme.py into the README. The cohort is reproduced by building g3-evolve at that commit and running each island with its seed, --h 0.4, --tier1-gate 62, --tier2-gate 71 and --log-every 50; pilot_tools export rebuilds the watertight envelope and print mesh from the genome row. The binary STL files (millimetres) are listed in Table 6.

9 Conclusion

Putting the edge and serration rules into the decoder made every design the search could express a knife with a correctly ground, double-serrated two-sided blade, and the search then

traded mass, strength and ergonomics within that space. The search also exposed a gap in the measurement chain: on its coarse grid, a design could look lighter than it would print. Re-measuring the ranking on the print grid closed that gap for this release. The best individual of the release, `hz4i013_g00995_d206e3c9`, scores 80.55 at 59.3 g with an ergonomic multiplier of 0.988. The limits in Section 7 set the next work: a stronger sharp point, a stiffer hollow grip under the spine load, a search grid fine enough that it cannot be gamed, and a longer cohort, all measured the same way.

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